

Empirical Proof Record: VALIDATED

2026-05-15T14:58:38.684512+00:00

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Proof ID: 69197dbcb7cc2e3a8ee46d2464d970f5...
Created: 2026-05-15T14:58:38.684512+00:00

1. Claim

Under sustained load on a balanced text corpus, Kimera’s entity target completes each cycle in under 4.00 seconds at the 95th percentile (the architectural throughput ceiling).

- **Operationalisation:** 95th-percentile of per-cycle wall-time (raw['cycle_seconds']) across the streamed batch; cycles that failed to record a cycle_seconds are excluded from the denominator
- **Threshold:** p95_cycle_wall_time_s <= 4.0 seconds
- **H0:** p95 wall-time > 4.00s
- **H1:** p95 wall-time <= 4.00s

2. Verdict

VALIDATED — observed 2.3572830475779467
measured per-cycle wall-time on 200/200 cycles (0 adapter errors); distribution: median 1.01s, p95 2.36s, p99 3.02s (range 0.08-3.26s); batch totals: wall 221.94s, cpu 243.86s (periodic_subprocess_sampler), rss_peak 4576.7MB, max threads 80, max procs 2

3. Evidence

Pillar	Statistic	Value	95th Percentile
0.engineering.p95_wall_time	p95_cycle_wall_time_s	2.3572830475779467	
0.engineering.wall_time_distribution	cycle_wall_time_distribution	1.0104102295008488	
0.engineering.batch_cpu	batch_cpu_total_s	243.85926604799997	
0.engineering.rss_peak	rss_peak_bytes	4576722944.0	

4. Pre-registration

- Registered at: 2026-05-15T14:54:56.599862+00:00
- Config hash: 1bfc2c127c850b1a04fdff3622986f3f...
- Data hash: 83109a27c3df2a45b912398b969fa8f7...

5. Signature

ae740c6246e190f06e71626ec8e618b199a6d54581640682254a1537ae02743a